

Problem

Design a T network necessary to realize the following z parameters at $\omega = 106 \text{ rad/s}$.

$$[z] = \begin{bmatrix} 4+j3 & 3 \\ 2 & 5-j \end{bmatrix} \text{ k}\Omega$$

Step-by-step solution

Step 1 of 3

Write the equation for a two-port network in the terms of the z parameters as follows:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \dots\dots (1)$$

Here,

V_1 is the voltage across port 1,

V_2 is the voltage across port 2,

I_1 is the current entering into the network through port 1,

I_2 is the current entering into the network through port 2,

z_{11} , z_{12} , z_{21} and z_{22} are the z parameters of the two-port network.

Step 2 of 3

Write the given z parameters matrix.

$$z = \begin{bmatrix} 4+j3 & 3 \\ 2 & 5-j \end{bmatrix} \text{ k}\Omega \dots\dots (2)$$

From (1) obtain the values of z_{11} , z_{12} , z_{21} and z_{22} .

$$z_{11} = (4+j3) \text{ k}\Omega$$

$$z_{12} = 3 \text{ k}\Omega$$

$$z_{21} = 2 \text{ k}\Omega$$

$$z_{22} = (5-j) \text{ k}\Omega$$

Since $z_{12} \neq z_{21}$, the network cannot be represented as T-network but can be represented by a more general equivalent circuit as shown in Figure 19.5(b) in the textbook.

Step 3 of 3

The general equivalent circuit is shown in Figure 1.

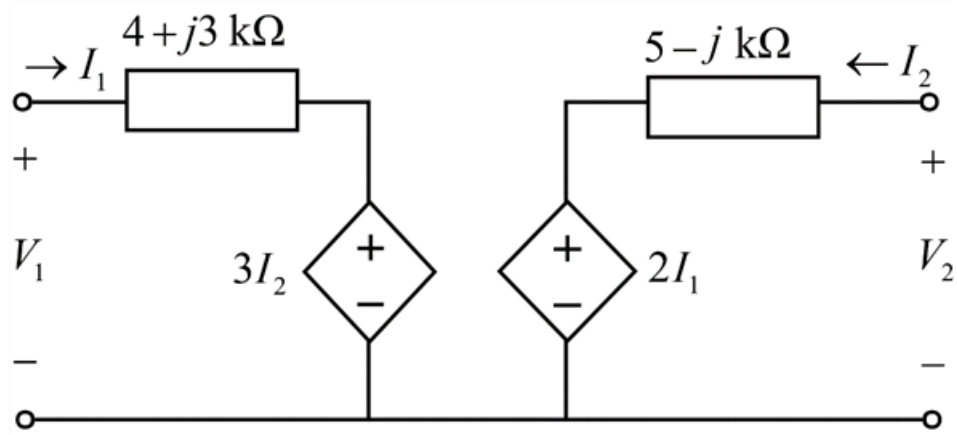


Figure 1

Therefore, as the network is not reciprocal, it cannot be represented as a T-network but a general equivalent network is shown in Figure 1.